

HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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ABSTRACT. A Hamiltonian bundle $M \hookrightarrow P \rightarrow X$ (with monotone compact fibers) induces a kind of “bundle of A_∞ categories” over X , with fiber given by the Fukaya category of M . Algebraic K -theory ideas, the above picture for $X = S^m$, and basic representation theory yield the following: if G is a compact Lie group and R is any commutative ring then there is a natural group homomorphism $\pi_m(BG) \rightarrow K_m^{\mathbb{Z}_2}(R)$. Here $K_m^{\mathbb{Z}_2}(R)$ is a variant of algebraic K -theory of R , isomorphic to $K_m(R) \oplus K_{m-1}(R)$ if R is a regular ring (e.g. $\mathbb{Z}$). Our construction naturally first assigns “Hamiltonian elements” in a certain categorified algebraic K -theory $K_m^{Cat}(R)$, analogous to Toën’s secondary K -theory of R . These gives a geometry powered proof that $K_2^{Cat}(\mathbb{Z})$ is infinitely generated (with details postponed). This is in contrast to finite generation of standard algebraic K -theory of $\mathbb{Z}$ already proved by Quillen. Taking the Langlands dual of G , we explore a conjectural relationship between the images of the corresponding homomorphisms above.

1. INTRODUCTION

A guiding idea is that a Hamiltonian bundle $M \hookrightarrow P \rightarrow Y$ gives rise to a kind of “bundle of A_∞ categories”, with fiber the Fukaya category of M . Precisely what this means is an interesting mathematical problem in its own right, and one answer is developed in [17]. Note that we are not talking about local systems, as the “bundles” that one gets are generally non-trivial over spheres as shown in [16].

The second idea, due to Toën [21], is that there is a refinement of K -theory of a commutative ring R , by replacing R -modules with dg or A_∞ categories over R , and using Waldhausen K -theory of a commutative ring. We show that these two ideas naturally fit together. We use the above assignment, with $Y = S^m$, to associate to each such P an element in categorified and standard algebraic K -theory of R . This culminates in a connection between symplectic geometry and algebraic K -theory, and in particular we get a geometric insight into algebraic K -theory of $\mathbb{Z}$, especially its categorified variant $K_m^{Cat}(\mathbb{Z})$. For example, we obtain an infinite generation result for $K_2^{Cat}(\mathbb{Z})$. At the end we explore connections with Langlands duality.

Although in the main examples here the principal actors are compact groups, in particular $PU(n)$, the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds; so we quickly introduce this. Let (M, ω) be a symplectic manifold, throughout assumed to be compact, with ω a closed nondegenerate 2-form on M . The group of diffeomorphisms ϕ of M , satisfying $\phi^*\omega = \omega$, is called the symplectomorphism group of (M, ω) . When M is simply connected, this is also the group of Hamiltonian symplectomorphisms, which we denote by $\mathcal{H} = \text{Ham}(M, \omega)$. In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields $\{X_t\}$, $t \in [0, 1]$, satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions $\{H_t\}$ on M .

The group $\mathcal{H}$ is one of the more enigmatic objects of mathematics. It is always (unless $M = pt$) an infinite-dimensional Fréchet Lie group with its C^∞ topology. But it has a natural, bi-invariant

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Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics, see for instance Polterovich [12].

Much of the algebraic/topological structure of $\mathcal{H}$ is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration* $M \hookrightarrow P \rightarrow Y$ is a smooth fiber bundle over Y , with fiber a symplectic manifold (M, ω) , whose structure group is contained in $\text{Ham}(M, \omega)$. For example, take $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$ where ω_{FS} is the Fubini-Study symplectic form, and take the structure group $G = PU(n+1) \subset \text{Ham}(\mathbb{CP}^n, \omega_{FS})$. We will from now on omit ω_{FS} from all notation, it is implicitly always the chosen symplectic form.

The other ingredient is algebraic K -theory, which is a construction associating an infinite loop space ¹ $K(R)$ to any ring R , and so abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called K -theory groups of R . This is a powerful tool in algebraic topology and ring theory, with applications for instance in number theory and geometric topology.

One motivating example for algebraic K -theory is the following. Take R to be the non-commutative ring $\mathcal{K}$ of compact operators on a separable complex Hilbert space, the algebraic K -theory space $K(\mathcal{K})$ ² is identified with the topological K -theory space $KU = BU \times \mathbb{Z}$, see Karoubi [6], Suslin-Wodzicki [20]. In particular, a complex vector bundle E over a space X determines a homotopy class $[f_E : X \rightarrow K(\mathcal{K})]$, which we might call a *geometric element* of the corresponding abelian group. We shall see that in a certain sense this extends to a general ring R , provided it is commutative, and replacing complex vector bundles by Hamiltonian fibrations.

For each commutative ring R , we construct “categorified algebraic K -theory groups” $K_m^{Cat}(R)$. These can be understood to be generated by aforementioned bundles of A_∞ categories over S^n with fibers A_∞ categories with finite type, see Definition 3.1. In principle, this categorified algebraic K -theory of R is a slight variation of the secondary K -theory of R introduced by Toën, with the main difference being that $\mathbb{Z}$ -graded chain complexes are replaced by $\mathbb{Z}_2$ -graded chain complexes. The following is a conjunction of Theorem 4.5 and Corollary 4.8.

Theorem 1.1. *If the Fukaya category of M is finite type, then for each R the data of the isomorphism class of $M \hookrightarrow P \rightarrow S^m$ determines an element in $K_m^{Cat}(R)$. This determines homomorphisms*

$$\phi_m : \pi_m(B\text{Ham}(\mathbb{CP}^n)) \rightarrow K_m^{Cat}(R),$$

and so homomorphisms

$$\phi_m : \pi_m(BPU(n)) \rightarrow K_m^{Cat}(R).$$

Example 1. In the case of $n = 1$, in [16], using geometric arguments we outline the proof of non-triviality of ϕ_4 for $R = \mathbb{Z}$. This contrasts with a theorem of Rognes [14] on the triviality of $K_4(\mathbb{Z})$.

Claim 1.2.

$$\phi_2 : \mathbb{Z}_p \simeq \pi_2(BPU(p)) \rightarrow K_2^{Cat}(\mathbb{Z})$$

is injective for each prime p . In particular, $K_2^{Cat}(\mathbb{Z})$ is infinitely generated.

Since the groups $K_m(\mathbb{Z})$ are always finitely generated by the foundational results of Quillen [13], the claim is perhaps surprising.

One geometric ingredient for the proof is the now classical part of symplectic geometry: the Seidel homomorphism [18]. The details will appear elsewhere.

¹In this paper, the term infinite loop space is meant to be synonymous with connective Ω -spectrum, i.e. a sequence of spaces $\{E_n\}_{n \in \mathbb{N}}$ with $E_n \simeq \Omega E_{n+1}$. Then $K(R)$ is the notation for the 0-space E_0 of the corresponding spectrum.

²Rather one must take a certain non-connective variant of the algebraic K theory construction.

There is a 2-periodic variant of algebraic K -theory of R , which we denote by $K_m^{\mathbb{Z}_2}(R)$. Technically, $K_m^{\mathbb{Z}_2}(R)$ is the algebraic K -theory space $K(W)$, where W is the Waldhausen category of 2-periodic complexes of R -modules.

Remark 1.3. It was pointed out to me by Bertrand Toën that for R a regular ring, for example $R = \mathbb{Z}$, $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$. This can be proved with A^1 homotopy theory. Although we don't technically need this.

The same geometric story yields the following.

Theorem 1.4. *For any compact Lie group G , there is a homotopy natural map:*

$$(1.5) \quad h : \mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R),$$

and so there are natural homomorphisms

$$h_m : \pi_m(\mathrm{BG}) \rightarrow K_m^{\mathbb{Z}_2}(R).$$

Taking the Langlands dual of G , we explore correspondence of the respective mappings in Section 6. In Section 4.1, we also outline how the maps h_m “lift” to $K_m^{\mathrm{Cat}}(R)$. To this end we need commonly expected monotonicity properties of certain Lagrangians associated to toric degenerations of G/T .

Question 1. Can one obtain the classes

$$[\mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R)]$$

by techniques of homotopy theory and algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

It should be noted that compared to the maps ϕ_m , showing non triviality of the maps h_m is harder (and is an open problem), even restricting to $PU(n)$ and $R = \mathbb{Z}$. This might seem paradoxical, but the idea is that $K_m^{\mathrm{Cat}}(R)$ sees more information, so it is easier to find nontrivial things in the image of ϕ_m .

2. PRELIMINARIES

We will mostly follow Keller [7], but our base commutative ring will be denoted R to avoid overloading notation. An A_∞ category C over R is a form of non-associative dg-category, with $\mathrm{hom}_C(a, b)$ a cochain complex over R . We refer the reader to standard references for further details, see for instance [11] which will come in use later.

Denote by $A_\infty \mathrm{Cat}_R^{\mathrm{strict}}$ the category of strictly unital, $\mathbb{Z}$ -graded A_∞ categories with morphisms strict unital A_∞ functors. We set $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$ to be like $A_\infty \mathrm{Cat}_R^{\mathrm{strict}}$ but with chain complexes $\mathbb{Z}_n$ -graded instead of $\mathbb{Z}$ -graded. This means that all structure maps, functors, etc., are with respect to the $\mathbb{Z}_n$ -grading. Various subscripts R may be omitted.

There is an endofunctor

$$T : A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}} \rightarrow A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}},$$

s.t. $T(B)$ is idempotent complete and pretriangulated for each B . We may denote by $\widehat{C}$ the category $T(C)$.

Let $\mathrm{Cat}_R^{\mathbb{Z}_n}$ denote the category of small R -linear $\mathbb{Z}_n$ -graded categories. For $C \in A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$, set $hC \in \mathrm{Cat}_R^{\mathbb{Z}_n}$ to be the category with the same objects and with $\mathrm{hom}_{hC}(a, b) = H^*(\mathrm{hom}_C(a, b))$.

Given a morphism $f : A \rightarrow B$ in $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$, we denote by $f_* : hA \rightarrow hB$ the induced morphism in $\mathrm{Cat}_R^{\mathbb{Z}_n}$. We say that f is a *quasi-equivalence* if f_* is an equivalence. We will say that f is a *Morita*

equivalence if the induced functor $T(A) \rightarrow T(B)$ is an equivalence. We denote by $Hmo^{\mathbb{Z}_2}$ the homotopy category of $A_\infty \text{Cat}_R^{\text{strict}, \mathbb{Z}_2}$ with respect to Morita equivalences, as explained by Keller this is a pointed category with the zero object: the category 0 with one object and zero endomorphism complex.

The category $A_\infty \text{Cat}_R^{\mathbb{Z}_n, \text{strict}}$ is cocomplete, see [11]. Furthermore, it has a model category structure [10]. This by left Bousfield localization yields a Morita model structure with weak equivalences Morita equivalences, although we do not explicitly need this model structure.

Definition 2.1. A *short exact sequence* in $A_\infty \text{Cat}_R^{\text{strict}, \mathbb{Z}_2}$ is a sequence $A \xrightarrow{I} B \xrightarrow{P} C$, such that in $Hmo^{\mathbb{Z}_2}$, I is the kernel of P and P is the cokernel of I .

As explained by Keller in [7], an explicit example for such a short exact sequence is given by a Drinfeld dg-quotient sequence $A \hookrightarrow B \rightarrow B/A$.

2.1. Simplicial sets. We will need basic theory of simplicial sets, particularly the notion of a simplex category, for the definition of bundles of A_∞ categories. We denote by Δ the simplex category:

- The set of objects of Δ is $\mathbb{N}$.
- $\text{hom}_\Delta(n, m)$ is the set of non-decreasing maps $[n] \rightarrow [m]$, where $[n] = \{0, 1, \dots, n\}$, with its natural order.

A simplicial set X is a functor

$$X : \Delta^{op} \rightarrow \text{Set}.$$

$X(n)$ is the set of n -simplices of X . Δ^d will denote a particular simplicial set: the standard representable d -simplex, with

$$\Delta^d(n) = \text{hom}_\Delta(n, d).$$

Morphisms of simplicial sets are given by natural transformations of functors and this forms a category denoted by $s\text{Set}$.

Definition 2.2. For X a simplicial set, $\Delta(X)$ will denote the *simplex category of X* . This is the category such that:

- The set of objects $\text{obj } \Delta(X)$ is the set of simplicial maps:

$$\Sigma : \Delta^d \rightarrow X, \quad d \geq 0.$$

- Morphisms $f : \Sigma_1 \rightarrow \Sigma_2$ are commutative diagrams in $s\text{Set}$:

$$(2.3) \quad \begin{array}{ccc} \Delta^d & \xrightarrow{\quad} & \Delta^n \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X. \end{array}$$

Given a morphism $f : X \rightarrow Y$ of simplicial sets, we set $\Delta f : \Delta(X) \rightarrow \Delta(Y)$ to be the functor defined on objects by $\Delta f(\Sigma) = f \circ \Sigma$. If $m : \Sigma_1 \rightarrow \Sigma_2$ is a morphism in $\Delta(X)$:

$$\begin{array}{ccc} \Delta^k & \xrightarrow{m} & \Delta^d \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X, \end{array}$$

then obviously the diagram below also commutes:

$$\begin{array}{ccc} \Delta^k & \xrightarrow{m} & \Delta^d \\ & \searrow h_1 & \downarrow h_2 \\ & & Y, \end{array}$$

where $h_i = \Delta f(\Sigma_i) = f \circ \Sigma_i$, $i = 1, 2$. And so the latter diagram determines a morphism $\Delta f(m) : h_1 \rightarrow h_2$ in $\Delta(Y)$. Clearly, this determines a functor Δf as needed.

Definition 2.4. Let X be a simplicial set, and C any category. Let $F_i : \Delta(X) \rightarrow C$, $i = 0, 1$ be functors. A **concordance** of F_0 to F_1 is a functor $\tilde{F} : \Delta(X \times I) \rightarrow C$ such that $\tilde{F}|_{\Delta(X \times \{0\})} = F_0$ and $\tilde{F}|_{\Delta(X \times \{1\})} = F_1$.

3. DEFINITION OF THE CATEGORIFIED ALGEBRAIC K-THEORY GROUPS OF R

We need to impose some algebraic finiteness conditions on our A_∞ categories, otherwise any K -theory we construct would be trivial due to the Eilenberg swindle, the same reason that K -theory of infinite dimensional vector bundles is trivial. A natural requirement is that our A_∞ categories be smooth and proper, or saturated in the sense of Kontsevich-Soibelman [9], as this is a dualizability condition, see [22]. In the $\mathbb{Z}_n$ -graded setting, the analogue of this condition is more complex, we will instead work with a less restrictive notion, which is not too hard to check for geometric examples.

Set $\mathcal{A} = \mathcal{A}_R \subset A_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$ to be the full subcategory whose objects are finite type, where the latter is defined as follows.

Definition 3.1. We say that $B \in A_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$ is **finite type** if the following conditions are satisfied.

- B is Morita equivalent to an A_∞ algebra, whose underlying chain complex is projective and finitely generated over R in each degree.
- The Hochschild homology complex of B over R is quasi-isomorphic (in the category of 2-periodic complexes over R) to a complex that is projective and finitely generated over R in each degree.

Set $\hat{\mathcal{A}}_R \subset \mathcal{A}_R$ to be the full subcategory with objects pretriangulated and idempotent complete. As outlined by Toën in [21], $\hat{\mathcal{A}}_R \subset \mathcal{A}_R$ has the structure of a Waldhausen category. The cofibrations are taken to be fully-faithful, strict, unital, injective on sets of objects A_∞ functors. Weak equivalences are Morita equivalences. To be strict this is not a Waldhausen category but ∞ -Waldhausen in the sense of Barwick [1], as for example the zero category 0 (the category with one object and zero complex as endomorphism set) is not initial in $\hat{\mathcal{A}}_R$ but the hom-sets $\text{hom}_{\hat{\mathcal{A}}_R}(0, A)$ are contractible when taken in the ∞ -localization of $\hat{\mathcal{A}}_R$ with respect to the Morita equivalences.

Given this we define

$$K_m^{\text{Cat}}(R) = \pi_m K(\hat{\mathcal{A}}_R),$$

where $K(\hat{\mathcal{A}}_R)$ is the associated Waldhausen spectrum. Calculations from this general perspective are difficult, we now describe a more concrete presentation.

3.1. Concrete presentation of $K_m^{\text{Cat}}(R)$.

Definition 3.2. Let X be a simplicial set. A **bundle of A_∞ categories over X** is a functor $F : \Delta(X) \rightarrow w\hat{\mathcal{A}}_R$. F will be called a **bundle functor over X** .

Suppose we have bundle functors over X , fitting into a sequence of natural transformations of functors to $\hat{\mathcal{A}}_R$ (as opposed to $w\hat{\mathcal{A}}_R$):

$$(3.3) \quad F_0 \rightarrow F_1 \rightarrow F_2,$$

such that for each object $\Sigma \in \Delta(S^k)$ the resulting sequence in $\mathcal{A}_R$ is a short exact sequence. Then we call $F_0 \rightarrow F_1 \rightarrow F_2$ a **short exact sequence**.

Definition 3.4. Let X be a Kan complex, Set $B(X, R)$ to be the free abelian group generated by the set of concordance classes $[F]$ of functors $F : \Delta(X) \rightarrow w\mathcal{A}_R$. And we define $K^{Cat}(X, R)$ to be the quotient of $B(X, R)$ by the subgroup generated by

$$\{[F_0] - [F_1] + [F_2] \mid \text{there is a short exact sequence } F_0 \rightarrow F_1 \rightarrow F_2\}.$$

This is clearly contravariantly functorial in X so if $f : X \rightarrow Y$ is a simplicial map then we get a group homomorphism $f^* : K^{Cat}(Y, R) \rightarrow K^{Cat}(X, R)$ defined on generators by $f^*([F]) = [F \circ \Delta f]$.

The groups $K^{Cat}(X, R)$ are covariantly functorial in R by “extension of scalars”. Furthermore, $X \mapsto K^{Cat}(X, R)$ gives a contravariant homotopy functor, as by definition of concordance it is clear that if $f : X \rightarrow Y$ is homotopic to g then $f^* = g^*$. In particular if X_0 is homotopy equivalent to X_1 then $K^{Cat}(X_0, R) = K^{Cat}(X_1, R)$.

Let $S_\bullet^n$ be the singular complex of S^n .

Definition 3.5. For $k > 0$, we define

$$K_n^{Cat}(R) = \ker\{(\Delta^0 \rightarrow S_\bullet^n)^* : K^{Cat}(S_\bullet^n, R) \rightarrow K^{Cat}(\Delta^0, R)\},$$

and define

$$K_0^{Cat}(R) = K^{Cat}(\Delta^0, R).$$

In other words, $K_0^{Cat}(R)$ is the quotient of the free group generated by Morita equivalence classes $[C]$, by the subgroup generated by:

$$\{[C_0] - [C_1] + [C_2] \mid \text{there is a short exact sequence } C_0 \rightarrow C_1 \rightarrow C_2\}.$$

To see relationship between the pair of definitions of $K_n^{Cat}(R)$, note that for X a Kan complex a bundle functor $F : |\Delta(X)| \rightarrow w\mathcal{A}_R$ induces a sequence

$$(3.6) \quad |X| \simeq |\Delta(X)| \xrightarrow{|F|} |w\mathcal{A}_R| \xrightarrow{i} K^{Cat}(R),$$

where $|X|$ also denotes the geometric realization of X . And clearly a concordance between F_0, F_1 induces a homotopy between $i \circ F_i$. Thus we get an induced map:

$$(3.7) \quad \tilde{\mathcal{J}}_X : B(X, R) \rightarrow Maps(|X|, K^{Cat}(R)),$$

where $hMaps(|X|, K^{Cat}(R))$ is the set of free homotopy classes $[|X| \rightarrow K^{Cat}(R)]$. As $K^{Cat}(R)$ is an infinite loop space, $hMaps(|X|, K^{Cat}(R))$ is naturally an abelian group.

Claim 3.8. *The map $\tilde{\mathcal{J}}_X$ passes to the quotient, inducing a map*

$$(3.9) \quad \mathcal{J}_X : K^{Cat}(X, R) \rightarrow hMaps(|X|, K^{Cat}(R)),$$

furthermore $\mathcal{J}_X$ is a group isomorphism, and induces a natural isomorphism $\mathcal{J}$ from the Ab-valued functor $X \mapsto K^{Cat}(X, R)$ to the functor $X \mapsto Maps(|X|, K^{Cat}(R))$.

The proof of the claim is postponed, as it involves various algebraic topology details. Note that in general one cannot always concretely present Waldhausen K -theory groups in this manner. The proof in our setting hinges on the fact that the ∞ -localization of $\hat{\mathcal{A}}_R$ with respect to Morita equivalences has a concrete model $\mathcal{M}$, such that concordance classes of bundle functors $F : \Delta(X) \rightarrow w\hat{\mathcal{A}}_R$ are in correspondence with homotopy classes $[X \rightarrow Core(\mathcal{M})]$, where $Core(\mathcal{M})$ is the maximal Kan subcomplex. As previously remarked, this $\mathcal{M}$ is the ∞ -Waldhausen category from which the spectrum $K^{Cat}(R)$ is constructed.

Given the claim, there is clearly an induced map

$$(3.10) \quad \mathcal{J}_m : K_m^{Cat}(R) \rightarrow \pi_m(K^{Cat}(R)),$$

and it is a group isomorphism.

3.2. A very basic computation. We recall the definition of $K_0(R)$. This is the quotient of the free abelian group generated by the set of isomorphism classes of projective finitely generated R -modules, by the subgroup generated by

$$\{[A] - [B] + [C] \mid \text{exists an exact sequence } 0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0.\}$$

If $\mathcal{E}$ is a triangulated category, one defines $K_0(\mathcal{E})$ to be the quotient of the free abelian group, generated by the set of isomorphism classes in $\mathcal{E}$, by the subgroup generated by

$$\{[A] - [B] + [C] \mid \text{exists a distinguished triangle } A \rightarrow B \rightarrow C \rightarrow A[1]\}.$$

It is known that $K_k(D^{perf}(Mod_R)) \simeq K_k(R)$, where Mod_R denotes the category of projective finitely generated R -modules and $D^{perf}(Mod_R)$ denotes the derived category of perfect complexes.

If $D_n(Mod_R)$ denotes the n -periodic derived category, we no longer have that $K_0(R) \simeq K_0(D_n(Mod_R))$, as there are certain corrections, see Saito [15]. However we have the following.

Theorem 3.11 (Saito [15]). *For n even, the natural homomorphism:*

$$(3.12) \quad \psi : K_0(D_n(Mod_R)) \rightarrow K_0(R)$$

$$(3.13) \quad \psi([V]) = \sum_{i=0}^{i=n} (-1)^i [H^i(V)],$$

is an isomorphism.

There is a group homomorphism

$$\phi : K_0^{Cat}(R) \rightarrow K_0(D_2(Mod_R))$$

induced by $[C] \mapsto [Hoch_\bullet(C)]$ for $Hoch_\bullet(C)$ the Hochschild homology complex of C . This readily follows from the fact that the Hochschild functor $Hoch_\bullet$ is “localizing” which means in this case that a short exact sequence

$$C_0 \rightarrow C_1 \rightarrow C_2$$

in $\mathcal{A}_R$ is taken to an exact triangle

$$Hoch_\bullet(C_0) \rightarrow Hoch_\bullet(C_1) \rightarrow Hoch_\bullet(C_2) \rightarrow Hoch_\bullet(C_0)[1],$$

in the derived category $D_2(Mod_R)$, see Keller [7, Theorem 5.2].

Corollary 3.14. *The composition map*

$$K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}})) \xrightarrow{\psi} K_0(\mathbb{Z}),$$

is non-trivial.

Proof. Take C to be a finite type $\mathbb{Z}_2$ -graded A_∞ algebra over $\mathbb{Z}$ such that $HH_0(C) = \mathbb{Z}$, $HH_1(C) = 0$, $HH_2(C) = \mathbb{Z}$, where $HH_\bullet(C)$ is the homology of $Hoch_\bullet(C)$. As a geometric example take $C = FH_\bullet(L_0, L_0)$ where the latter is the Floer cochain algebra of the equator L_0 in S^2 , then the Hochschild homology $HH_\bullet(C)$ is isomorphic to the singular homology $H_\bullet(S^2, \mathbb{Z})$, [19].

So we get

$$\psi([Hoch_\bullet(C)]) = 2[\mathbb{Z}] \neq 0 \in K_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

□

A further corollary of the proof above is:

Corollary 3.15. $[Fuk(S^2, \omega)]$ represents a non-trivial element in $K_0^{Cat}(\mathbb{Z})$.

One natural question is how to produce interesting elements in $K_m^{Cat}(R)$ for higher m . We will construct such elements using Hamiltonian fibrations and Floer theory.

4. BUNDLES OF A_∞ CATEGORIES FROM HAMILTONIAN FIBRATIONS

The main construction of [17] has input a smooth Hamiltonian fibration $M \hookrightarrow P \rightarrow Y$, with (M, ω) a compact monotone symplectic manifold, e.g. $\mathbb{CP}^n$. More generally, a monotone symplectic manifold (M, ω) is one satisfying $[\omega] = \text{const} \cdot c_1(TM)$, with $\text{const} > 0$, as functionals $\pi_2(M) \rightarrow \mathbb{R}$.

Denote by $Y_\bullet$ the simplicial set of smooth maps $\Delta^n \rightarrow Y$ for Δ^n also denoting the standard topological n -simplex.

The output is a functor

$$(4.1) \quad F_P : \Delta(Y_\bullet) \rightarrow wA_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}},^3$$

that is a bundle functor over $Y_\bullet$, whose concordance class F_P is independent of auxiliary choices (these mainly consist of choices of almost complex structures and Hamiltonian connections.) For $x : \Delta^0 \rightarrow Y$ a point map $F_P(x) = \text{Fuk}(P_x, \omega_x)$, P_x denoting the fiber over x .

We can outline this further, if $\Sigma : \Delta^1 \rightarrow Y$ is a path from x to y , then $F_P(\Sigma)$ has the set objects

$$\text{obj } F_P(x) \sqcup \text{obj } F_P(y).$$

For $L_x \in \text{obj } F_P(x)$ and $L_y \in \text{obj } F_P(y)$, $\text{hom}_{F_P(\Sigma)}(L_x, L_y)$ is a certain chain complex generated by flat sections of an auxiliary Hamiltonian connection on Σ^*P , with end points of the sections on L_x and L_y .

More generally, for a smooth $\Sigma : \Delta^n \rightarrow Y$ denote by $v_i : \Delta^0 \rightarrow Y$, $0 \leq i \leq n$, the corresponding vertex maps. Then $F_P(\Sigma)$ has the set of objects

$$(4.2) \quad \sqcup_i \text{obj } F_P(\Sigma)(v_i).$$

The hom sets are defined analogously so that an edge inclusion morphism $\Sigma^1 \rightarrow \Sigma$ in $\Delta(Y_\bullet)$ induces a fully-faithful embedding $F_P(\Sigma_1) \rightarrow F_P(\Sigma)$. Here by an edge inclusion we mean a morphism of the type:

$$(4.3) \quad \begin{array}{ccc} \Delta^1 & \xrightarrow{\quad} & \Delta^n \\ & \searrow \Sigma^1 & \downarrow \Sigma_2 \\ & & X. \end{array}$$

The only deep part of the construction is then the definition of the A_∞ structure on $F_P(\Sigma)$. This involves some novel geometric topology of the universal curve, but we will not attempt to outline this further, instead referring the reader to the original construction in [17].

Definition 4.4. We call a compact monotone symplectic manifold *finite monotone type*, if its monotone Fukaya category (see [19] and also [17]) is finite type.

Let us set $\widehat{F}_P = T \circ F_P$. We then have the following corollary of [17, Theorem 6.6].

Theorem 4.5. *Let $M \hookrightarrow P \rightarrow Y$ be a Hamiltonian fibration with fiber (M, ω) finite monotone type. Then the assignment:*

$$(4.6) \quad [P] \mapsto [\widetilde{\mathcal{J}}_\bullet(\widehat{F}_P)] \in h\text{Maps}(Y, K^{\text{Cat}}(R)),$$

is well defined, where $[P]$ is the Hamiltonian isomorphism class of the fibration P , and $\widetilde{\mathcal{J}}$ is as in (3.7).

As an example, we have:

³In [17] we work with $R = \mathbb{Q}$ but this was only used for inverting quasi equivalences. As we don't need to do this here, the construction works over any commutative ring.

Proposition 4.7. $\mathbb{CP}^n$ is finite monotone type.

Proof. Specifically, we take a concrete model of $\text{Fuk}(\mathbb{CP}^n)$ with the set of objects given by the orbit of the Clifford torus L_0 by the natural $\text{Ham}(\mathbb{CP}^n, \omega)$ action. In this case, it is automatic that $\text{Fuk}(\mathbb{CP}^n)$ is quasi-isomorphic to the A_∞ algebra $\text{hom}_{\text{Fuk}(\mathbb{CP}^n)}(L_0, L_0)$. The Hochschild chain complex of the latter is quasi-isomorphic to the Hamiltonian Floer chain complex, see for instance [19], which in each degree is finitely and freely generated over $\mathbb{Z}$. And so $\text{Fuk}(\mathbb{CP}^n)$ is finite type for any R . $\square$

In particular, taking $Y = S^m$ we get:

Corollary 4.8. For each commutative ring R , there are natural homomorphisms:

$$(4.9) \quad \phi_m : \pi_m(B\text{Ham}(\mathbb{CP}^n)) \rightarrow K_m^{\text{Cat}}(R).$$

And in particular natural homomorphisms:

$$(4.10) \quad \phi_m : \pi_m(BPU(n)) \rightarrow K_m^{\text{Cat}}(R).$$

Proof. Let $\mathbb{CP}^n \hookrightarrow P \rightarrow B\text{Ham}(\mathbb{CP}^n)$ be the tautological Hamiltonian fibration, that is the fiber bundle associated to the universal principal $\text{Ham}(\mathbb{CP}^n)$ bundle over $B\text{Ham}(\mathbb{CP}^n)$. Use the conjunction of Proposition 4.7 and Theorem 4.5 to get set maps ϕ_m . To see that they are homomorphisms note that we have a sequence of maps well defined up to homotopy:

$$B\text{Ham}(\mathbb{CP}^n) \xrightarrow{|\widehat{F}|} |w\widehat{\mathcal{A}}_R| \xrightarrow{i} K^{\text{Cat}}(R),$$

where

$$F : \Delta(B\text{Ham}(\mathbb{CP}^n)_\bullet) \rightarrow w\mathcal{A}_R,$$

is the bundle functor associated to the universal Hamiltonian $\mathbb{CP}^n$ fibration over $B\text{Ham}(\mathbb{CP}^n)$. By the naturality of the assignment $[P] \mapsto [F_P]$, see [17, Section 9.2], ϕ_m are the induced maps

$$(i \circ |F|)_* : \pi_m(B\text{Ham}(\mathbb{CP}^n)) \rightarrow \pi_m(K^{\text{Cat}}(R)).$$

And so are group homomorphisms. $\square$

4.1. Generalizing to G/T . Given a compact Lie group G , the maximal coadjoint orbit G/T can be made monotone with respect to the Kirillov-Kostant-Souriau symplectic structure ω_{KKS} , which for G/T is uniquely determined up to deformation equivalence, see for instance [8]. We will from now on omit this monotone ω_{KKS} from notation, but implicitly it will always be the chosen symplectic form. If $G = PU(n)$ there is a distinguished monotone Lagrangian torus L_{GC} of $PU(n)/T$, arising as the central fiber of a Gelfand-Cetlin system, see [3, Theorem B]. This is a kind of analogue of the Clifford torus.

It is natural to expect that $\text{Fuk}(G/T)$ is finite type. In this case we similarly get homomorphisms $\pi_m(B\text{Ham}(G/T)) \rightarrow K_m^{\text{Cat}}(R)$. Note that when we talk about $\text{Fuk}(G/T)$, we are talking about a model ⁴ with finitely many objects up to the $\text{Ham}(G/T)$ action, similarly to how we treated $\text{Fuk}(\mathbb{CP}^n)$.

For a concrete example, form the subcategory $C_{L_{GC}}$ of $\text{Fuk}(PU(n)/T)$ with objects formed by the $\text{Ham}(PU(n)/T)$ -orbit of L_{GC} . In this case the finite type condition is forced and we get natural homomorphisms

$$\phi_{L_{GC}, m} : \pi_m(B\text{Ham}(PU(n)/T)) \rightarrow K_m^{\text{Cat}}(R),$$

by the same construction; that is getting bundles of A_∞ categories over S^m with fiber $C_{L_{GC}}$. More explicitly, given a fibration $PU(n)/T \hookrightarrow P \rightarrow S^m$, and $x : \Delta^0 \rightarrow Y$ a point map, set $F_P^{\mathcal{L}_{GC}}(x) = C_{\mathcal{L}, x}$

⁴Not necessarily Morita equivalent to $\text{Fuk}(G/T)$.

where $C_{\mathcal{L}_{GC}} \subset \text{Fuk}(P_x, \omega)$, (identifying P_x with $PU(n)/T$ up to $\text{Ham}(PU(n)/T)$ action). For a general Σ , $F_P^{\mathcal{L}_{GC}}(\Sigma)$ is a category with finitely many objects, all of which are isomorphic to a distinguished object, corresponding to $\mathcal{L}_{GC}$; this follows by the prescription in (4.2).

If we restrict to $G = PU(n)$, we get two maps $\phi_{L_{GC}, m} : \pi_m(BPU(n)) \rightarrow K_m^{Cat}(R)$, and $\phi_m : \pi_m(BPU(n)) \rightarrow K_m^{Cat}(R)$. These are likely distinct, they correspond to taking different coadjoint orbits of G , and so different fibrations over $BPU(n)$.

We can also do this for G/T (and more general coadjoint orbits) by constructing a distinguished Lagrangian $\mathcal{L}$, as the central fiber of a toric degeneration of G/T . One of the most well known setups for this is via so called string polytopes and Newton–Okounkov bodies see [4], and G/T always has a toric degeneration. For a general G it is expected that $\mathcal{L}$ is monotone, (this in principle depends on the chosen degeneration, but it is enough for us, if we just want some map to K -theory, that exists one degeneration, such that $\mathcal{L}$ is monotone). For example, this holds for $PU(n)/T$ as in this case $\mathcal{L}$ can be identified with L_{GC} for the toric degeneration associated to the Gelfand–Cetlin polytope.

Assuming this for the moment we get a finite type subcategory $C_{\mathcal{L}} \subset \text{Fuk}(G/T)$, with the set of objects the orbit of $\mathcal{L}$ by the $\text{Ham}(G/T)$ action. If we take this $C_{\mathcal{L}}$ as our model we again get maps:

$$(4.11) \quad \phi_{\mathcal{L}} : \pi_m(B\text{Ham}(G/T)) \rightarrow K_m^{Cat}(R).$$

5. HAMILTONIAN ELEMENTS IN ALGEBRAIC K -THEORY

One of the goals is to obtain “Hamiltonian elements” in classical algebraic K -theory of R . Let $M \hookrightarrow P \rightarrow Y$ be a Hamiltonian fibration, with compact monotone fiber (M, ω) . Let

$$F_P : \Delta(Y_{\bullet}) \rightarrow wA_{\infty}Cat_R^{\mathbb{Z}_2, strict}$$

be as in the previous section. Then the composition: $\Phi_P = Hoch_{\bullet} \circ F_P$ gives a functor to $wCh_R^{\mathbb{Z}_2}$, where $Ch_R^{\mathbb{Z}_2}$ is the Waldhausen (strict) category $Ch_R^{\mathbb{Z}_2}$ of 2-periodic complexes of projective finite rank R -modules. Its Waldhausen K -theory infinite loop space will be denoted by $K^{\mathbb{Z}_2}(R)$.

Restrict again to $Y = B\text{Ham}(G/T)$, and P the associated G/T bundle over $B\text{Ham}(G/T)$. Now by standard homotopy theory, $|\Delta(Y_{\bullet})| \simeq X$, where for a category C , $|C|$ is shorthand for the geometric realization of the nerve. Then we get a map:

$$|F_P| : B\text{Ham}(G/T) \rightarrow |wCh_R^{\mathbb{Z}_2}|,$$

whose homotopy class is well defined by the fact that the concordance class of F_P is well defined.

There is a natural map $|wCh_R^{\mathbb{Z}_2}| \rightarrow K^{\mathbb{Z}_2}(R)$, by generalities of the Waldhausen construction, see [23, Remark 8.3.2]. Composing with this map we get a natural homotopy class:

$$(5.1) \quad [|F_P| : B\text{Ham}(G/T) \rightarrow K^{\mathbb{Z}_2}(R)].$$

Proof of Theorem 1.4. Compose the natural map

$$BG \rightarrow B\text{Ham}(G/T),$$

and the map from (5.1) to obtain h . □

6. LANGLANDS DUAL ALGEBRAIC K -THEORY ELEMENTS.

It is natural to ask how Langlands duality fits into our story. Denote by $K^{\vee}$ the maximal compact subgroup of the Langlands dual $G_{\mathbb{C}}^{\vee}$ of the complexification $G_{\mathbb{C}}$. Via Theorem 1.4, we have natural maps:

$$\begin{aligned} h_{G, m} &: \pi_m(BG) \rightarrow K_m^{\mathbb{Z}_2}(R), \\ h_{K^{\vee}, m} &: \pi_m(BK^{\vee}) \rightarrow K_m^{\mathbb{Z}_2}(R). \end{aligned}$$

If G is simply laced (e.g. $G = SU(n)$) it is not hard to see that the images of these maps are the same. In this case $G/T = K^\vee/T$ and then the images of G and $K^\vee$ in $\text{Ham}(G/T)$ coincide, so that the assertion readily follows. So we may propose a conjecture :

Conjecture 1. *For each compact Lie group G the images of $h_{G,m}$ and $h_{K^\vee,m}$ coincide.*

Furthermore, there should be “lifts” as in (4.11):

$$\begin{aligned}\phi_{\mathcal{L},G} : \pi_m(BG) &\rightarrow K_m^{\text{Cat}}(R), \\ \phi_{\mathcal{L},K^\vee} : \pi_m(BK^\vee) &\rightarrow K_m^{\text{Cat}}(R),\end{aligned}$$

and we can again ask if the images coincide. For $m = 0$, this is saying that $Fuk(G/T)$ and $Fuk(K^\vee/T)$ ⁵ represent the same classes in $K_0^{\text{Cat}}(R)$. This assertion holds if these categories were Morita equivalent, but it is much weaker. The case $m = 0$ is then loosely comparable to the homological mirror symmetry conjecture of Kontsevich.

The above is thematically connected to topological aspects of Langlands duality/correspondence as in [2] for example. But I don’t know a direct mathematical connection. From a physical perspective of Kapustin-Witten [5], geometric/topological aspects of Langlands correspondence are expressions of the electric-magnetic duality. It would be good to understand if there are likewise some physical connotations of the maps $h_{G,m}$.

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⁵Or to be more restrictive their models $C_{\mathcal{L}}$ as in Section 4.1.

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